

No.	Solution	Remarks
7(a)(i)	The number of ion pairs formed is given by the area under the line in Fig. 7.2. Accept: Calculation/estimation of the area under the line.	[1]
7(a)(ii)	Reasoning: By conservation of energy, the energy to produce each ion-pair is converted from the kinetic energy of the α-particle. Kinetic energy of α-particle $= 240000 \times 30$ $= 7.2 \times 10^6 \text{ eV}$ $= 7.2 \text{ MeV}$	[1] for correct reasoning [1] for correct answer
7(a)(iii)	Beta particle is <u>less ionising</u>, i.e. fewer ionisations per unit distance travelled. Hence, <u>it travels further before all its energy transferred</u>. Accept: Beta particle <u>interacts less frequently with air molecules due to its smaller size</u>. Hence, it <u>travels further before all its energy transferred</u>.	[1]
7(b)(i)	${}_{88}^{224}\text{Ra} \rightarrow {}_{86}^{220}\text{Ra} + {}_2^4\alpha$	[1]
7(b)(ii)	 By principle of conservation of linear momentum, $\begin{aligned} + \quad 220u \quad v_R + 4u \quad -v_\alpha &= 0 \\ \rightarrow \end{aligned}$ $\frac{v_\alpha}{v_R} = \frac{220}{4}$ $= 55$	[1] for correct $\frac{v_\alpha}{v_R}$

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	$\frac{KE_{\text{total}}}{KE_{\alpha}} = \frac{\frac{1}{2} 4u v_{\alpha}^2 + \frac{1}{2} 220u v_R^2}{\frac{1}{2} 4u v_{\alpha}^2}$ $= 1 + 55 \left(\frac{1}{55} \right)^2$ $= \frac{56}{55}$ Hence, proportion of total energy the α-particle carries $= \frac{KE_{\alpha}}{KE_{\text{total}}} \times 100\%$ $= \frac{55}{56} \times 100\%$ $= 98.2$ $\approx 98\%$	[1] for correct $\frac{KE_{\text{total}}}{KE_{\alpha}}$ [1] for correct proportion %
7(c)(i)	Mass difference $\Delta m = m_{\text{Ra}} - m_{\text{Rn}} + m_{\alpha}$ $= [223.97191 - 219.96420 + 4.00150] \text{ u}$ $= 0.00621 \text{ u}$ Using $\Delta E = \Delta mc^2$, Total energy released in a single decay $\Delta E = 0.00621 \times 1.66 \times 10^{-27} \times 3.00 \times 10^8 \text{ }^2$ $= 9.278 \times 10^{-13}$ $\approx 9.28 \times 10^{-13} \text{ J}$ Energy of an α-particle $E_{\alpha} = 0.98 \times 9.278 \times 10^{-13}$ $= 9.092 \times 10^{-13}$ $\approx 9.09 \times 10^{-13} \text{ J}$	[1] for correct mass difference [1] for energy released in a single decay [1] for energy of an α-particle
7(c)(ii)	Total number of decays during the therapy of duration of 2 h $= \text{activity} \times \text{time}$ $= 18 \times 10^3 \times 2 \times 60 \times 60$ $= 1.296 \times 10^8$ $\approx 1.3 \times 10^8 \text{ (shown)}$ Assumption: Activity of radioactive source remains constant over two-hour period of the therapy.	[1] for correct calculation and showing calculated answer before 1.3×10^8 [1] for correct assumption

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7(c)(iii)1.	Total energy absorbed by the tissue $= 9.092 \times 10^{-13} \times 1.3 \times 10^8$ $= 1.182 \times 10^{-4}$ $\approx 1.18 \times 10^{-4} \text{ J}$	[1] for correct answer
7(c)(iii)2.	Absorbed dose to the tissue $= \frac{\text{total energy absorbed by the tissue}}{\text{mass of tissue}}$ $= \frac{1.182 \times 10^{-4}}{4.0 \times 10^{-3}}$ $= 0.02955$ $\approx 0.0296 \text{ Gy}$	[1] for correct expression and numerical substitution [1] for correct answer
7(c)(iii)3.	Equivalent dose to the tissue $= QF \times \text{absorbed dose}$ $= 20 \times 0.02955$ $= 0.591$ $\approx 0.59 \text{ Sv}$	[1] for correct answer
7(d)(i)	Using $A = \lambda N$, Number of iodine-131 nuclei, $N = \frac{A}{\lambda} = \frac{A}{\frac{\ln 2}{t_{1/2}}}$ $= \frac{3.7 \times 10^9}{\ln 2} \times 8.04 \times 24 \times 60 \times 60$ $= 3.708 \times 10^{15}$ $\approx 3.71 \times 10^{15}$	[1] for correct expression and numerical substitution [1] for correct answer
7(d)(ii)	Iodine-123 emits only γ-ray photons, whereas iodine-131 emits both β-particles and γ-ray photons. Hence, making iodine-123 safer and more suited for imaging due to the reduced potential harm to healthy tissues. Iodine-131's beta emissions are more useful for therapeutic rather than imaging purposes Accept: 1. Iodine-123 half-life is shorter than that of iodine-131, which is long enough to conduct imaging but short enough to minimise radiation burden on the body tissues. Iodine-131 has a longer half-life (about 8 days), leading to prolonged radiation exposure. 2. Iodine-123 emits gamma rays of energy ($\sim 0.16 \text{ MeV}$) that are ideal for detection by gamma cameras, resulting in clearer images. Iodine-131 emits higher-energy gamma rays ($\sim 0.36 \text{ MeV}$), which are less optimal for imaging and can	[1]

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	cause image blurring due to Compton scattering within the tissues.	